Computer Laboratory Surveillance System: Robbery Scene Detection and Alerting

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***Abstract*—This paper presents the smart surveillance system using YoLo v3 to detect the robbery scene and send an alert notification through Line application. Our system is capable of detecting the moving objects (CPUs and monitors) and people in computer laboratory. The surveillance system can recognize the authorized persons and captures the suspect who try to steal laboratory's device (system assumption: an unauthorized person is carrying CPU and a monitor out of the laboratory). We used YoLo v3 for CPUs and monitors detection, face detection and recognition, and intersection over union (IOU) technique for the bounding box. The sensitivity, specificity, and accuracy of our system are 0.72, 0.64, and 0.68, respectively.**

Keywords*—*Object detection, Face detection, Face recognition, YoLo v3, IOU

I. Introduction

Many schools provide facilities and equipments for students to practice, e.g., Physics, Chemistry, and Computer laboratory. Some devices especially in a computer laboratory are expensive and can be moved, e.g., CPU, monitor, and printer. Monitoring these devices by installing a surveillance system or school securities cannot guarantee that they would not be stolen. This paper presents the practical smart surveillance system to detect robbery scenes and send an alert notification through Line application. YoLo v3 and IOU were used for object detection and tracking. The system is capable of recognizing the authorized persons' face; whereas captures the robbery scene by assuming that someone else is taking a device out of the laboratory. The system was implemented and tested in the real place (Department of Computer and Information Science, Faculty of Applied Science, KMUTNB). The system was evaluated using sensitivity, specificity, and accuracy. The paper is organized into 5 parts: introduction, preliminaries, methods, experimental results, and conclusion.

II. Preliminaries

CCTV (Closed-circuit television) can be divided into 5 types [1]: 1) the standard camera, this type of camera suits for the place which has light through; the standard camera might be installed within the camera housing for outdoor usage. 2) infrared camera, this type of camera can be used in the dark place (indoor and outdoor usage). 3) dome camera, this type of camera can be rotated in 360 degrees and suits for the place which has light through (same as the standard camera). 4) speed dome camera, the enhanced features, e.g., pan, tilt, and zoom are added in dome camera; more option for speed dome camera is it can be controlled remotely by using the control box. 5) hidden camera or pin camera, this spy camera can be installed in a tiny place; normally, this is an illegal use because of the privacy concerns. Figure 1 shows 5 types of CCTV (This work was implemented by using dome camera).

Fig. 1. Different types of CCTV.

We investigated many papers about object detection and we focused on YoLo. YoLo (You Only Look Once) [2], is the state-of-the-art, real-time object detection algorithm. The latest release of YoLo is version 5. The recent application of using YoLo is proposed in [1], [3]; the work implemented YoLo v4 to detect the pistol to reduce crimes such as shoplifting, robbery, and rape. In [4], YoLo is proposed for front yard surveillance to recognize the robbery activities. The system could detect human and car in the front yard, then the computer vision algorithm will determine the pattern of human movement that seems to be robbery. In [5], the paper presented deep learning and machine learning techniques to

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observe the real-time ATM robbery. The system will send the alert when a thief is about to break the ATM machine or destroy the ATM camera. In [6], the paper proposed computer vision-based robbery alerting system. The system can detect armed theft wearing full and partial mask using CCTV cameras, the contribution of the work is YoLo object detection model and image processing techniques, e.g., filtering, and facial landmark identification.

III. METHODS

This paper used YoLo v3 as an object detection. YoLo is capable of detecting multiple objects in the same scene. In this work, the surveillance must detect people, CPUs, and monitors at the same time. When the detection has been established, the convolutional neural network (CNN) will segment all objects in the scene into bounding boxes. Each bounding box will be covered all objects in the scene, YoLo will estimate the number of bounding boxes and it will choose the bounding boxes which are the best fit to the objects (Figure 2).

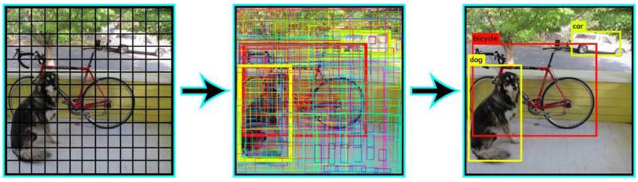

Fig. 2. YoLo algorithm [7].

YoLo v3 uses 53 layers of CNN, called Darknet-53. In the object detection task, we need to double these 53 layers to 106 layers. The 106 layers of CNN in YoLo v3 is shown in Figure 3.

	Type	Filters	Size	Output
	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
1×	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
	Convolutional	128	3 × 3 / 2	64 × 64
2×	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
	Convolutional	256	3 × 3 / 2	32 × 32
8×	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
8×	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
4×	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Fig. 3. YoLo v3 architecture.

IOU for bounding box: IOU or Jaccard Similarity Index is the statistical technique used to measure the similarity of 2 objects based on set theory. In image processing, we can represent all pixels in the digital image as a set. Therefore, we can use set operations to each pixel in an image [8], [9]. Let P and G are set of pixels, contained in the bounding box. IOU can be defined in equation 1:

$$IOU\ (P, G) = \frac{|P \cap G|}{|P \cup G|} \quad (1)$$

We illustrate this equation in Figure 4. We can compute the area of:

$P \cap G$ = (20-12)x(12-8) = 8x4 = 32

$P \cup G$ = The area of blue bounding box + The area of orange bounding box - $P \cap G$, thus

$P \cup G$ = 10x7 + 13x12 – 32 = 70 + 156 – 32 = 194, thus

$IOU\ (P, G)$ = 32/194

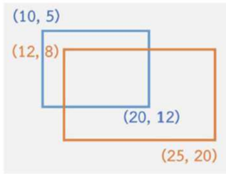

Fig. 4. $IOU\ (P, G)$.

To train the YoLo, we recorded a robbery scene by demonstrating both of laboratory staff and theft try to carry computer devices out of a room. For a non-robbery scene, we recorded a laboratory staff and theft carrying computer devices within a room. We split a video file into consecutive frames for our training dataset. We prepared the images of laboratory staff by using 80 facial images and 400 images of computer devices in the different positions for training the YoLo object detection (Figure 5). We also designed web application for the client side as the GUI of the system, and the server side for connecting CCTV and database management.

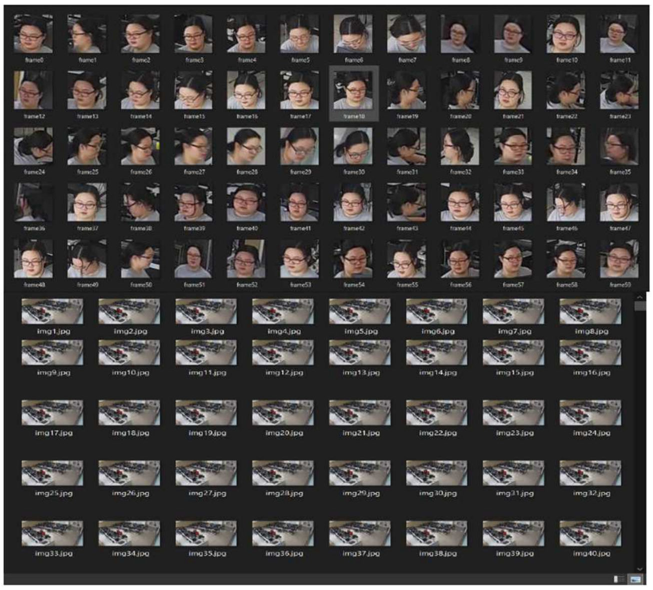

Fig. 5. Images used in our experiment.

We used LabelImg for those images which were mentioned earlier. The results of LabelImg (YoLo reformatting) consist of 2 files: weight and cfg (Figure 6).

Fig. 6. YoLo v3 files (.cfg and .weight).

The surveillance system architecture is shown in Figure 7 and the flow chart of the system is shown in Figure 8, respectively. The video data will be stored to the database through the server side. The user/system administrator can access the web application by their roles (client side). When the system detects an unauthorized person trying to carry CPU/monitor out of the laboratory, the system will send an alert notification to Line application (the robbery scene captured image will be sent via Line message).

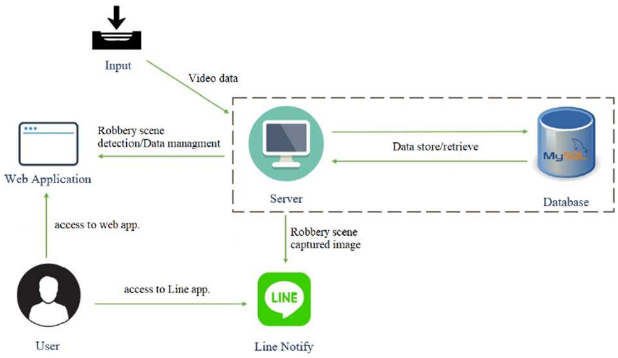

Fig. 7. The system architecture.

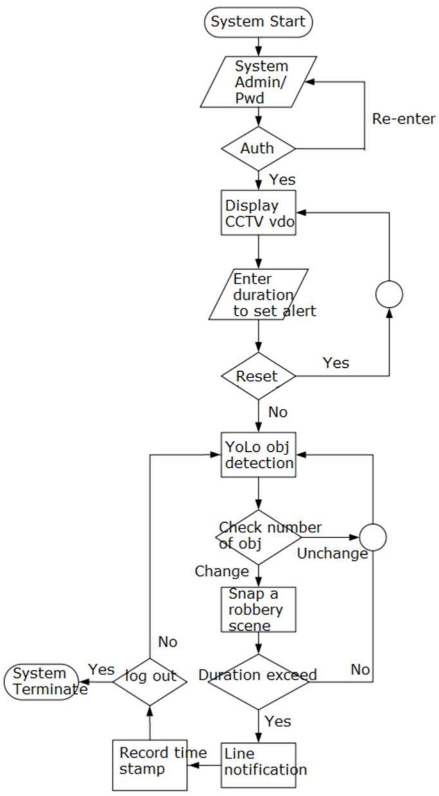

Fig. 8. The system flowchart.

IV. Results

The results of robbery detection are shown in Figure 9. Figure 9b, the system detected that human face coming in the laboratory and recognizes that she was a laboratory staff. Figure 9d, the system detected that a laboratory staff was carrying a monitor out of the laboratory, and there was no alert notification. Figure 9e and 9f, the system detected that an unauthorized person was trying to move a monitor, and she carried a monitor out of the laboratory. Figure 9g, the system sent the robbery scene and notified the staff/administrator via Line application.

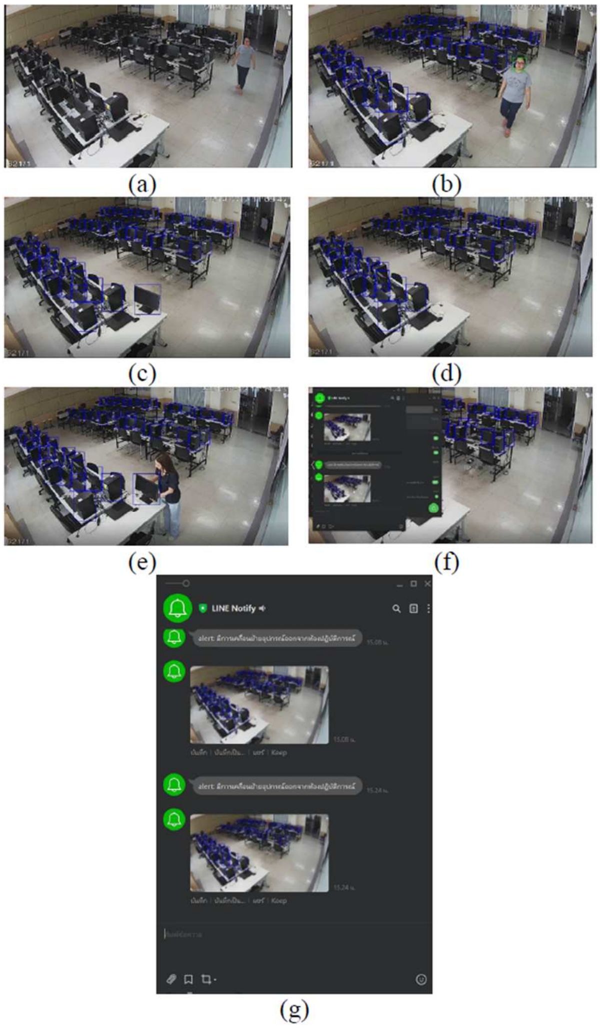

Fig. 9. The detection of robbery scene.

The GUI of the system is shown in Figure 10. The administrator can log in to the system and choose the date which he/she would like to see the video data. The video data according to the selection will be shown on the screen.

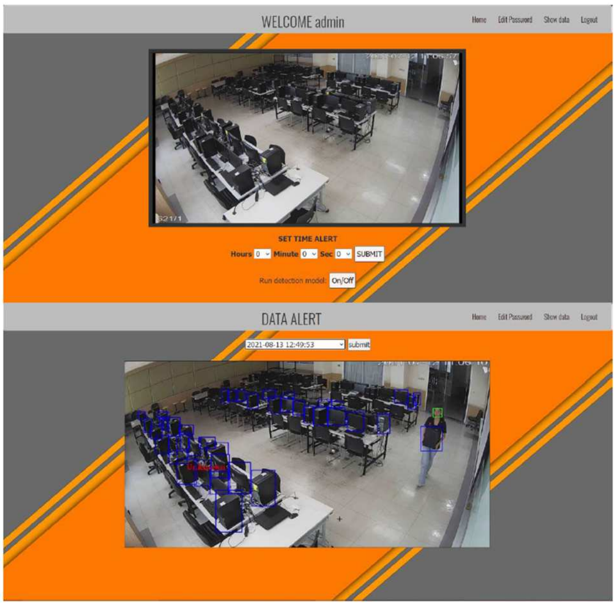

Fig. 10. The GUI of the system.

We evaluated our experiment using sensitivity, specificity, and accuracy [10]. According to *TP* (true positive), *TN* (true negative), *FP* (false positive), and *FN* (false negative); derived equations are shown below:

$$Sensitivity = \frac{TP}{TP+FN} \tag{2}$$

$$Specificity = \frac{TN}{TN+FP} \tag{3}$$

$$Accuracy = \frac{TP+TN}{TN+TP+FN+FP} \tag{4}$$

Table 1 and 2 show the experimental results, the sensitivity, specificity, and accuracy of our system are 0.72, 0.64, and 0.68, respectively.

TABLE I. NO ROBBERY SITUATION

Sample	Assumption: No robbery set up		
	Number of test	TN	FP
Laboratory staff 1	25	15	10
Laboratory staff 2	25	17	8
Total	50	32	18

TABLE II. ROBBERY SITUATION

Sample	Assumption: Real robbery set up		
	Number of test	TN	FP
Theft suspect 1	25	16	9
Theft suspect 2	25	20	5
Total	50	36	14

V. CONCLUSION

In this paper, the smart surveillance system is developed to automatically detect robbery that happens in the computer laboratory. The surveillance system is developed using YoLo v3, face detection and recognition techniques, and IOU for bounding box (object tracking). This work aims to help the school to protect the properties from the theft in real time by flag an alarm when robbery is detected. The system will send the robbery scene and notify to Line application. The limitation of this work is a poor number of camera and the environmental setting e.g., there is only 1 camera in a room and the layout of computer tables and chairs is not appropriate (there are some blind spots of a camera) . Therefore, in the 1st phase of this work, we must do our work in this circumstance. A future improvement of this work will focus on more sophisticate computer visions techniques and real time object detection using multiple CCTV cameras.

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